

Software Test Plan (STP)

Traffic Management System (Simulation)

Field	Details
Project	Traffic Management System (Simulation)
Team	2
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1. Introduction

Purpose: This document defines the test plan for the Traffic Management System (Simulation) v1.0. It outlines testing objectives, scope, strategy, resources, schedule, responsibilities, test metrics, and traceability for the simulation.

Scope: Testing covers vehicle movement, road and lane management, traffic-density sensing, adaptive traffic signal control, congestion handling, incident handling, emergency vehicle priority, simulation control, data logging, and performance of the simulated traffic environment. Physical traffic lights, real roadside hardware, and live city traffic infrastructure are excluded.

References: Traffic Management System SRS v1.0, Software Architecture and Design Specification v1.0, UML design diagrams, test cases, and requirements traceability information.

Definitions: TMS (Traffic Management System), STP (Software Test Plan), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UI (User Interface), KPI (Key Performance Indicator).

2. Test Items

- Traffic simulation engine
- Vehicle and traffic-flow module
- Road and lane management module
- Traffic sensor / density detection module
- Adaptive traffic signal controller
- Incident and emergency vehicle management module
- Simulation dashboard / user interface

- Logging and reporting module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- TMS-F-001: Create and initialize the traffic simulation
- TMS-F-002: Generate and move vehicles according to road and lane rules
- TMS-F-003: Detect vehicle count / traffic density using simulated sensors
- TMS-F-004: Change traffic signal states according to configured timing
- TMS-F-005: Adapt signal timing based on traffic density
- TMS-F-006: Detect and manage traffic incidents
- TMS-F-007: Give priority to emergency vehicles during an incident
- TMS-F-008: Display traffic status, signal state, congestion, and incidents
- TMS-F-009: Record simulation events and traffic statistics
- TMS-NF-001: Simulation response/update time within the defined performance threshold
- TMS-NF-002: Stable operation during the planned simulation load

4. Features Not to be Tested

- Physical traffic signal hardware and roadside sensor hardware
- Real-world vehicle positioning and GPS accuracy
- Live municipal traffic-control infrastructure
- Real payment, toll, or external transportation-system integrations
- Production deployment and city-scale field validation

5. Test Approach / Strategy

Levels:

- Unit tests (module-level logic such as vehicle movement, density calculation, and signal-state transitions)
- Integration tests (sensor → controller → signal and incident → priority-signal workflows)
- System tests (end-to-end simulation scenarios)
- Acceptance tests (user validation of the simulation dashboard and required scenarios)

Types:

- Functional testing of traffic and signal-control features
- Regression testing after changes to simulation logic
- Performance testing under increasing vehicle and event loads
- Usability testing of the simulation dashboard
- Boundary and negative testing for invalid configuration values and abnormal traffic conditions

Entry Criteria: Stable build delivered, requirements and test data available, simulation environment configured, and required test scenarios identified.

Exit Criteria: 100% of planned test cases executed, no critical defects open, major defects have an approved resolution plan, and all acceptance criteria for the simulation are satisfied.

5.1 Security Validation

- Validate that only authorized simulation users can modify control parameters where authentication is implemented.
- Validate input handling for vehicle counts, signal timings, road configuration, and incident parameters.
- Verify that logs do not expose unnecessary user or system-sensitive information.
- Test resistance to malformed or unexpected simulation input.
- Validate access control for administrative simulation functions.

6. Test Environment

Hardware: Standard development/testing computer capable of running the traffic simulation and visualization.

Software: Traffic Management System simulation build, configured road network, vehicle models, signal controller, and simulation dashboard.

Tools: Project implementation language tools, unit-test framework, UI testing tools where applicable, Git for version control, and Jira or an equivalent defect tracker.

Test Data: Different vehicle-density levels, road/lane configurations, signal timings, incident types, incident severities, and emergency-vehicle scenarios.

7. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated where applicable)
- Test Scripts
- Test Data and simulation scenarios
- Test Execution Logs
- Defect Reports
- Test Summary Report

8. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Project Team	Prepare plan, coordinate test execution, review coverage
Test Engineer	Project Team	Design and execute test cases, record results and defects
Developer	Project Team	Support defect fixes, debugging, and technical triage
Product Owner	Project Team	Review acceptance criteria and approve readiness

9. Risks and Mitigation

Risk	Mitigation
Simulation build instability	Use smoke tests before full execution and maintain a known stable build.
Incorrect traffic-model assumptions	Use multiple traffic-density and road-network scenarios and document assumptions.
High simulation load reduces performance	Run controlled load tests and optimize vehicle/signal update loops.
Defects in signal-state transitions	Use state-transition tests, boundary cases, and automated regression tests.

10. Assumptions & Dependencies

- The simulation build and required dependencies will be available before execution.
- Road, lane, vehicle, sensor, and signal configuration data will be provided.
- Expected behavior and acceptance criteria will be available from the SRS/design documents.
- The test machine will have sufficient resources to execute the planned simulation scenarios.
- Defect fixes will be delivered in a testable build for regression testing.

11. Suspension & Resumption Criteria

Suspend testing if:

- The simulation environment is unavailable or cannot execute planned scenarios.
- The build is unstable enough to block more than 30% of planned test cases.
- A critical defect prevents meaningful execution of core traffic-control workflows.

Resume testing if:

- Blocking defects are resolved or an approved workaround is available.
- The simulation environment is stabilized and test data can be loaded correctly.
- A verified build is provided for the affected test scenarios.

12. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. The following representative mapping is used for the simulation:

Requirement	Representative Test Cases
TMS-F-001 (Simulation initialization)	TC-SIM-01, TC-SIM-02
TMS-F-002 (Vehicle movement)	TC-VEH-01, TC-VEH-02
TMS-F-003 (Traffic-density detection)	TC-SEN-01, TC-SEN-02
TMS-F-004 (Signal state control)	TC-SIG-01, TC-SIG-02
TMS-F-005 (Adaptive signal timing)	TC-ADP-01, TC-ADP-02
TMS-F-006 (Incident handling)	TC-INC-01, TC-INC-02
TMS-F-007 (Emergency priority)	TC-EMG-01, TC-EMG-02
TMS-NF-001 (Performance)	TC-PERF-01, TC-PERF-02
TMS-F-008 (Dashboard/status display)	TC-UI-01, TC-UI-02
TMS-F-009 (Event/statistics logging)	TC-LOG-01, TC-LOG-02
TMS-NF-002 (Stable operation under planned load)	TC-REL-01, TC-REL-02

14. Test Metrics & Reporting

Metrics collected:

- % of planned test cases executed
- % passed / failed / blocked
- Defect density and severity distribution
- Defect aging
- Requirement coverage
- Average simulation update/response time
- Vehicle throughput and congestion-related KPI values where applicable

Reports:

- Daily execution status
- Defect status report
- Requirement coverage / RTM report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Project Team	16/09/2026
Dev Lead	Project Team	16/09/2026
Product Owner	Project Team	16/09/2026